AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 98

TARO 3 x 3 PT coupled to KUBOTA RK70-S Diesel Engine

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TARO 3 x 3 PT Centrifugal Pump coupled to a KUBOTA RK70-S Diesel Engine

Test Report No.	:	2013 - 98
Date of Test	:	May 29, 2013
Test Valid Until	:	May 29, 2018
Brand	:	TARO
Model	:	3 x 3 PT
Serial No.	:	No data
Type and Size	:	Single-end suction, 75 x 75 mm, Self-Priming Centrifugal pump
Manufacturer	:	LI KUANG MECHANICAL & INDUSTRIAL WORKS CO., LTD. 7 Lane 433 Chung Hsiao Rd., Chia Yi City, Taiwan
Test Requested by	:	KUBOTA PHILIPPINES, INC. 232 Quirino Highway, Baesa, Quezon City, Philippines
Machine Classification	:	Commercial

SUMMARY

The TARO 3 x 3 PT centrifugal pump coupled to KUBOTA RK70-S diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump used double groove B-section pulleys both with actual diameter of 101.6 mm (4.0 inches) and two pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 19.3 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2450 and 2556 rpm, respectively. The Maximum Engine Fuel Consumption Rate was 1.44 L/h. The Maximum Discharge Capacity of 8.02 L/s occurred at 2.40 m Total Head with Fuel Consumption Rate of 1.42 L/h. The shaft speed of the pump and engine at

this point were 2419 and 2525 rpm respectively. The Maximum System Efficiency of 6.19% occurred at 5.05 L/s Discharge Capacity and 15.6 m Total Head.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 7.51 m. The Net Positive Suction Head Available at this point was 2.29 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1.**

The test was conducted on May 29, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.

Figure 1. TARO 3 x 3 PT Centrifugal pump coupled to KUBOTA RK70-S Diesel engine on the test set-up

METHOD OF TEST

The pumpset was tested following the "Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test".

DESCRIPTION OF THE PUMP

The TARO 3 x 3 PT Centrifugal pump coupled to KUBOTA RK70-S Diesel Engine shown in **Figure 2**, was a single-end suction, 75 mm x 75 mm, centrifugal pump coupled to a 5.22 kW (7.0 Hp) KUBOTA RK70-S diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the engine and pump pulleys were both 101.6 mm (4.0 inches). The suction inlet and discharge outlet were threaded and designed for a 75 mm diameter pipe. The impeller was a semi-open type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing. The priming inlet was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a packing seal. The pump shaft rotated on ball bearing kept in place by a bearing housing integral to the casing.

The primemover, a KUBOTA RK70-S Diesel engine, was a 5.22 kW (7.0 hp), four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized an oil bath type air cleaner. Lubrication system was force-feed type. Cooling was done through the circulation of water from the cylinder jacket to the radiator through the thermo-siphon principle. It was started by a hand crank system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.

Figure 2. TARO 3 x 3 PT Centrifugal Pump coupled to a KUBOTA RK70-S Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the TARO 3 x 3 PT Centrifugal Pump coupled to KUBOTA RK70-S Diesel Engine

1.	Brand		:	TARO
2.	Model		:	3 x 3 PT
3.	Serial No.		:	No data
4.	Make		:	LI KUANG MECHANICAL & INDUSTRIAL WORKS CO., LTD. 7 Lane 433 Chung Hsiao Rd., Chia Yi City, Taiwan
5.	Type		:	Single-end suction, self priming Centrifugal pump
6.	Size (mm)		:	75 x 75
7.	Overall Dimension and Weight (Including mounting frame and engine)			
	7.1	Length (mm)	:	1160
	7.2	Width (mm)	:	505
	7.3	Height (mm)	:	575
	7.4	Weight (kg)	:	138.0
8.	Impeller			
	8.1	Type	:	Closed
	8.2	Diameter (mm)	:	176
	8.3	Width (mm)	:	16
	8.4	No. of vanes	:	8
	8.5	Material	:	Brass
9.	Suction Inlet			
	9.1	Type	:	Threaded flange
	9.2	Diameter (mm)	:	75
	9.3	Material	:	Cast Iron

Table 1. (Cont'n)

10.	Discharge Outlet		
	10.1 Type	:	Threaded flange
	10.2 Diameter (mm)	:	75
	10.3 Material	:	Cast Iron
11.	Casing		
	11.1 Type	:	Volute
	11.2 Material	:	Cast Iron
12	Stuffing box	:	Packing Seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	1800 - 2200
15.	Rated Maximum Discharge Capacity (L/s)	:	21.7
16.	Rated Maximum Total Head (m)	:	25.0
17.	Primemover (based on Plate Rating)		
	17.1 Brand	:	KUBOTA
	17.2 Model	:	RK70-S
	17.3 Serial No.	:	RK70-ACW3966
	17.4 Make	:	Indonesia
	17.5 Type	:	Single cylinder, 4-stroke cycle, Water-cooled, single horizontal diesel engine
	17.6 Rated Output Power (kW)		
	Maximum	:	5.22 @ 2400 rpm
	Continuous	:	4.48 @ 2200 rpm

Table 1. (Cont'n)

17.7	Bore x Stroke (mm)	:	80 x 75
17.8	Displacement Volume (cc)	:	376
17.9	Cooling System	:	Water-cooled
17.10	Lubricating System	:	Force-feed
17.11	Starting System	:	Rope-recoil
17.12	Air Cleaner	:	Oil bath type
17.13	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested : TARO 3 x 3 PT Centrifugal pump Coupled to KUBOTA RK70-S diesel engine

Date of Test : May 29, 2013

Test Condition

Ambient Temperature (^oC) : 32.2
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (^oC) : 33.5

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP-TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
2556	2450	1.05	19.3	0	0	11.0	0
2547	2445	1.23	17.9	3.91	0.68	12.8	5.34
2541	2438	1.20	15.6	5.05	0.78	15.5	6.19
2538	2436	1.35	14.2	5.78	0.81	14.0	5.75
2533	2429	1.37	12.8	6.33	0.80	14.3	5.58
2532	2423	1.42	11.5	6.65	0.75	14.7	5.10
2529	2422	1.43	10.0	6.89	0.68	14.9	4.56
2527	2419	1.42	8.60	7.09	0.60	14.7	4.07
2527	2419	1.36	7.50	7.24	0.53	14.2	3.75
2525	2419	1.44	4.50	7.66	0.34	15.0	2.24
2525	2419	1.42	2.40	8.02	0.19	14.8	1.28

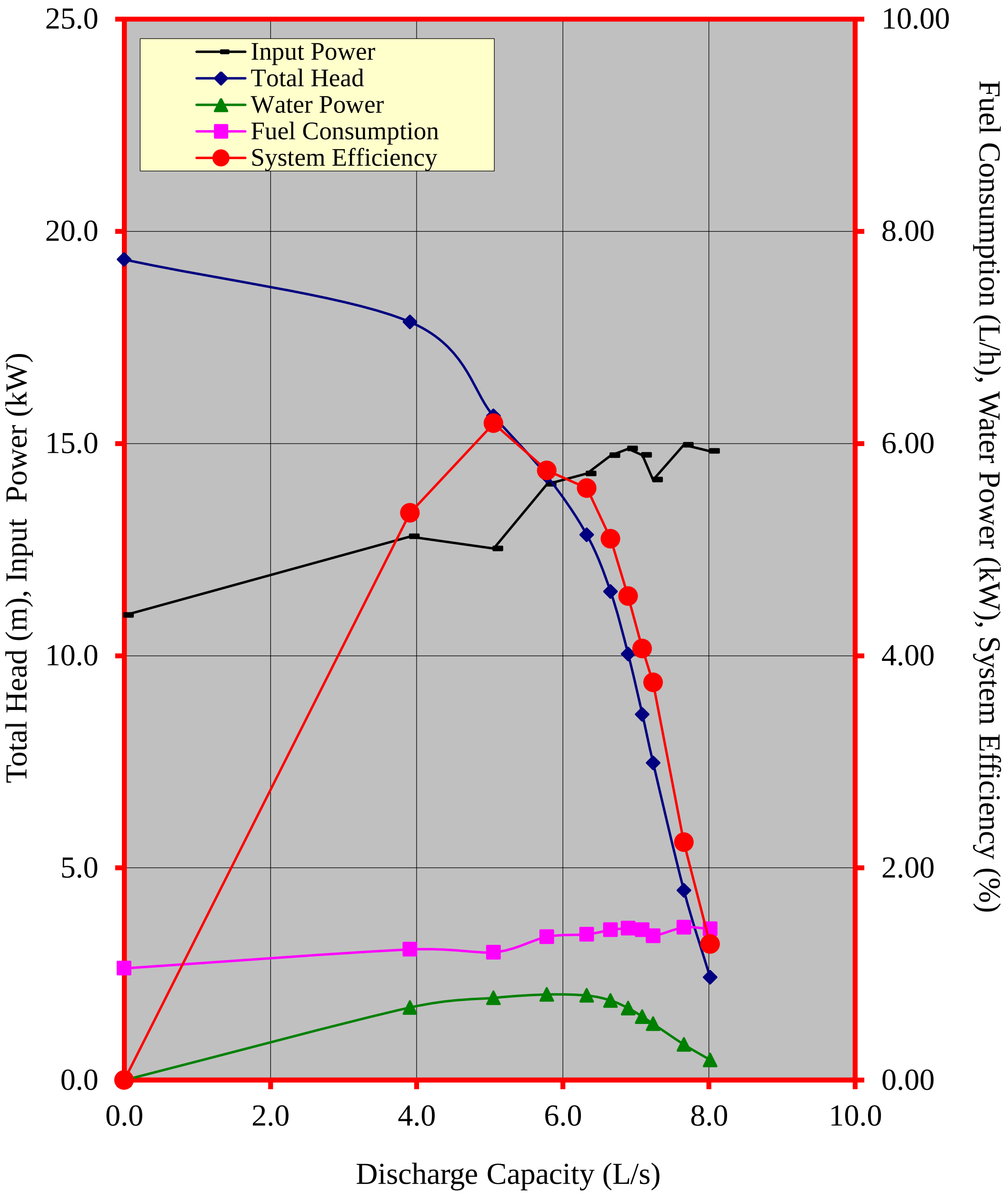

Figure 2. Performance characteristic curves of the TARO 3 x 3 PT pump coupled to KUBOTA RK70-S Diesel Engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested : TARO 3 x 3 PT Centrifugal pump coupled to KUBOTA RK70-S diesel engine

Date of Test : May 29, 2013

Test Condition

Ambient Temperature (^{o}C) : 32.5
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (^{o}C) : 33.5

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
2527	2420	0.55	9.25	27.3	8.07
2532	2424	1.80	8.0	26.3	7.18
2534	2428	3.20	6.60	24.4	5.98
2535	2429	4.59	5.21	23.6	4.79
2540	2431	5.98	3.82	23.0	3.53
2540	2435	6.67	3.13	21.7	2.59
2540	2431	7.37	2.43	20.3	0.99
2541	2435	7.51	2.29	18.4	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

Where: $$H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted to meter. Computed by determining the velocity of liquid on the pipe by formula, V = Q/A where V, velocity of liquid inside the pipe in meter per per second, Q, Discharge rate of the pump in liters per second, and A, Cross-sectional area of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity,Q, in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.

3. Water Power,WP, is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \, X \, Q}{102}$$

4. Input Power , IP, is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determine by the formula:

$$IP = \frac{T \, X \, N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters
P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

Test Engineer:

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N.B.

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